



A Corneal Surface Reflections-Based Intelligent System for Lifelogging Applications

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ABSTRACT

Corneal Surface Reflections, or reflections on our eye-surface, have been shown as a valid and more socially acceptable source of information for passive lifelogging applications by prior work. However, automatic analysis of corneal surface reflections from a single RGB camera to support passive lifelogging is not extensively investigated in prior work. To address this, we developed a synthetic and self-supervised learning-based two-stage pipeline of deep learning models to detect objects in these reflections. Our prototype only consists a single RGB camera looking into the eye. We collected data from different users in uncontrolled environments using the prototype and trained our system to detect multiple classes of objects present in a typical office environment. We then evaluated our model in partially-controlled and in-the-wild scenarios. In addition, based on the findings from a follow up user study and prior work, we discuss strengths and weaknesses of our system and using corneal surface reflections for passive lifelogging. Finally, we open-source our source codes and trained checkpoints.

1. Introduction

The cornea of the human eye acts as a mirror and reflects light from the environment. These reflections may contain identifiable objects and can be extracted from an image of the eye. Hence, a number of research works (El Hafi et al., 2016, 2017a, 2017b; Lander et al., 2016; Lander & Krüger, 2018; Nitschke et al., 2013) have utilized corneal images for lifelogging purposes. Their devices use cameras pointing towards eyes, making them more socially acceptable than other systems that require outward-pointing cameras (El Hafi et al., 2017a; Lander & Krüger, 2018).

Lander et al. were the first to explore the feasibility of lifelogging using corneal images. Preliminary results of a later work (Lander & Krüger, 2018) showed the potential of using neural networks to detect object in eye reflections. El Hafi et al. used a combination of a mathematical 3D eye model and a deep learning model to determine gaze and recognize objects in corneal surface reflections. However, it demands a calibration process and requires users to hold objects close to their eyes. Such a setting is not highly feasible for daily lifelogging. Our previous work (Kaluarachchi et al., 2021) used a similar setup to detect the presence of a screen in the corneal surface reflections along with determining cognitive load from the eye movements to build a tool for HCI researchers. However, there we have only used corneal surface reflections for binary classification.

In this work, we aim to evaluate the use of automatic corneal surface reflections analysis for lifelogging that works

for multiple people with multiple classes without having to calibrate for each individual. For this, we use a visual feed from a single RGB camera looking towards the eye, which would eventually contribute towards wearable lifelogging devices that are more socially acceptable (El Hafi et al., 2017a). Firstly we collect corneal images from multiple people without controlling any environmental or user characteristics-related parameters. Secondly, we developed a pipeline of deep learning algorithms to process these images. For the corneal image collection, we developed a prototype (see Figure 1) similar to prior works (Kaluarachchi et al., 2021; Lander & Krüger, 2018; Lander et al., 2016), which is based on a commercially available eye-tracking device¹. This device possesses a single web camera pointing towards an eye and can constantly capture corneal images. Compared to many of the prior works, our technique does not use typical infrared-based gaze tracking. Instead, our system just crops the reflective area of the eye directly using an object detection model on RGB images.

There are several challenges for developing a system for automatic corneal surface reflections analysis. According to our preliminary experiments, reflection of the target object accounts for less than 5% of the total pixels. This is mainly due the convex nature of the eye surface. As such, due to the extremely low signal-to-noise ratio, use of the entire image as the input is extremely inefficient. Also, the corneal reflection images collected in uncontrolled environmental settings can be noisy and heterogeneous (see Figure 2 for illustrated explanations). Training a deep learning model on



Figure 1. Prototype device in front (left) and side (right) views.

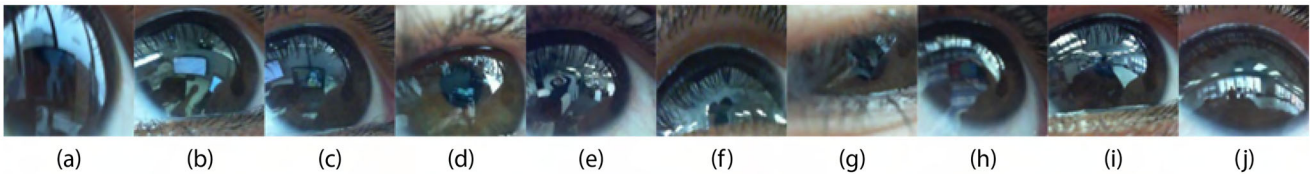


Figure 2. The figure shows some of the challenging situations due to the uncontrolled environment settings and eye characteristics. Images (a) and (b) show similar type screens appear in different shapes (warped) due to the angle of the cornea. Images (c) and (d) are both laptop screens but appear in different sizes due to the distance from the eye. Images (e) and (f) both show two different crops of a person in the reflection in two different poses due to the angle the person is looking. Images (g) and (h) both contain mobile screens but hard to identify due to the amount the person has opened the eye and the screen's content. Image (i) contains a person with whom the person wearing the device was having a chat, while image (j) shows a person far away who happened to be in the frame. In both cases, the person appears far and hard to classify into two classes.

this dataset from scratch requires a significant amount of annotated data to reduce model variance. This constitutes another challenge because it is prohibitively expensive to obtain such annotation.

In this article, we present an AI approach to tackle the above challenges. For the first challenge of signal-to-noise ratio, we designed a region of interest (ROI, i.e., where the reflections present) detection mechanism to crop the interested area. To build a small network for improved embedded systems compatibility, we develop and train a custom Convolutional Neural Network (CNN) with synthetic image-pretraining. To address the second challenge of limited annotated data, we use self-supervised (Kolesnikov et al., 2019a) pretraining that leverages the knowledge acquired by processing a large amount of unlabeled data.

Following the Human-Centered Machine Learning approach Kaluarachchi et al. (2021); Margetis et al. (2021); Shneiderman (2022a, 2022b), we evaluated our system with the collected annotated data by conducting leave-one-user-out analysis with 9 people. The results show that our system can achieve more than 75% accuracy for 5 classes of objects in a generic office environment, which is a significant improvement compared to the reported 70% single class accuracy of our prior work (T. I. Kaluarachchi et al., 2021).

We also present an analysis for different class variations to understand confusions among classes and to evaluate the stability of our training. In addition, in a follow-up user study we investigated the in-the-wild performance and discuss the opportunities and challenges of using our approach in a practical real-world lifelogging application.

In summary, our contributions are two-fold:

- Evaluation with multiple people and multiple classes to validate that corneal surface reflections from a single RGB camera can be used for lifelogging wearable devices
 - Results of multiple class-variations with leave-one-out settings.
 - Investigation of the model performance in an in-the-wild context and its implications for lifelogging application development
 - Discussion of design implications and possible future improvements based on end-user feedback.
- Making our work available to the research community to further validate and build upon
 - Our labelled dataset is available² upon request for research purposes.
 - We open sourced³ all our source codes, especially the SSL pretrained and finetuned checkpoints of our neural networks.

2. Related work

2.1. Object recognition in corneal surface reflections

There has been a number of work for recognizing objects in corneal surface reflections. Early attempts mainly employed mathematical models on non-distorted images. For instance, Nishino and Nayar (2006) first investigated the corneal surface reflections towards understanding the world surrounding the user. Nitschke and Nakazawa (2012) have attempted to improve the resolution of the corneal surface reflections with a mathematical model for eye tracking and pupil cropping. A group of researchers (Takemura et al., 2014) used a 3D eye tracking model with a similar head-mounted web camera setup to detect the focused object. Guestrin and Eizenman (2006) have used point reflections on the eye in infrared images from fixed infrared light sources to estimate gaze. Other works tried to separate reflections from the eyes to analyze illumination levels using traditional computer vision models (Wang et al., 2005, 2008).

Recently, several studies attempted to conduct object recognition on distorted images. El Hafi et al. tried to recognize objects in eye reflections with a mathematical eye model and a deep learning model. There have been attempts to measure human attention using corneal surface reflections (Lander et al., 2017). While the eye-tracking model makes several assumptions and requires precise calibration, they only showed results for a single person. There are works (Jenkins & Kerr, 2013) developed on classifying the faces appear on cornea reflections highlighting the importance of such systems in crime investigations. In our most recent work, EyeKnowYou T. I. Kaluarachchi et al. (2021), we built a deep learning-based toolkit to recognize screen and cognitive load using corneal surface reflections and eye movements. We used two binary classifiers for screens and cognitive load. Built on top of this work, here we explored recognition of various object classes to support lifelogging applications.

Although many of the mentioned works tried to recognize objects in corneal surface reflections, few of them specifically investigated lifelogging. Lander et al. first explored the concept of using corneal surface reflections for lifelogging. They conducted a manual analysis of objects in reflections as a feasibility study. This was later followed by another work (Lander & Krüger, 2018) to compare manual and automatic analysis of corneal surface reflections. Their automatic analysis utilized a YOLO (Redmon & Farhadi, 2018) model trained on the generic Imagenet (Russakovsky et al., 2015) dataset. However, they had not finetuned the network on a data distribution consisting corneal images and they highlight the importance of training on domain specific data. Apart from these, the most related work to this is an attempt to use an Inception V3 (Szegedy et al., 2015) model to detect hand-held objects and they have highlighted the problem of having to calibrate the eye model from user to user. However, none of the existing works have investigated beyond a single person which is a vital exploration to create practical solutions for lifelogging. Commercial products use infrared cameras pointing inwards

and RGB world cameras pointing outwards for eye-tracking and object recognition. Outward-pointing world cameras have social acceptance issues (El Hafi et al., 2017a; Lander & Krüger, 2018). In contrast to these prior work, we investigate the possibility of using corneal surface reflections beyond a single person, and we also open-source our trained models and the checkpoints for future work.

2.2. Deep learning for object detection and recognition

Majority of the modern day visual information processing algorithms use Convolutional Neural Networks (CNN) which is the current state-of-the-art for that. CNNs came to existence after (*Neocognitron*) was introduced (Fukushima, 1980) as the first CNN architecture, but without a parameter optimization method. Following the revolutionary invention of *Back Propagation* (Rumelhart et al., 1988), Back Propagation was successfully used for parameter optimization of CNNs (LeCun et al., 1989) to recognize hand written zip codes. With the introduction of AlexNet (Krizhevsky et al., 2012) to win the ImageNet (Russakovsky et al., 2015) challenge with a CNN, CNNs gained popularity exponentially followed by many innovations such as ZFNet (Zeiler & Fergus, 2013), VGG (Simonyan & Zisserman, 2014), GoogLeNet (Szegedy et al., 2015), Inception Net (Szegedy et al., 2015), and ResNet (He et al., 2016). Apart from these object recognition networks, object detection networks such as YOLO (Redmon & Farhadi, 2018), SSD (W. Liu et al., 2016), and RetinaNet (Lin et al., 2017) have been developed in parallel. However, deep learning is growing exponentially and newer and better algorithms arrive each year. Top performing model on Imagenet to the date for object recognition is an EfficientNet-L2 trained using the Meta Pseudo Labels (Pham et al., 2020) mechanism, which a semi-supervised learning scheme. Interestingly, transformer (Dosovitskiy et al., 2020; Vaswani et al., 2017) based Swin Transformer (Liu et al., 2021)—does not use CNN architecture—is the top performer for COCO (Lin et al., 2014) dataset, which is a famous benchmark for object detection.

However, it is important to note that all these benchmarking datasets have enormous amounts of labeled data making most of these challenges supervised learning challenges. For instance Imagenet has 14 million labeled images of 1000 classes. However, most of the real world problems like corneal surface reflections do not have large labeled data making it a challenging problem to tackle. Therefore we leverage a state-of-the-art technique called self-supervised learning to tackle the low data problem we have at hand.

2.3. Self Supervised Learning (SSL) in computer vision

Self Supervised Learning is a type of representation learning that aims to learn how to represent a given data distribution with lower dimensions. The specialty of SSL is that it can do it without having annotated data using a certain set of tasks called pretext tasks. For image data distributions, these tasks can be as simple as predicting if an image is upright or

rotated by 90° , 180° , or 270° . Some of such tasks are rotation recognition (Gidaris et al., 2018), solving a 3×3 jigsaw puzzle (Kim et al., 2018), and determining a relative patch location (Doersch et al., 2015) and identifying if two modified image segments are from the same image or different images. Labels for these tasks can be automatically generated (self-generated) while the training is happening. This is the main reason for identifying this as self-supervised rather than unsupervised. These pretext tasks help the network learn the general nature of the data distribution which can be later used as the starting point for a downstream task like object classification. This can be done in two ways. The first one is to use the SSL-trained network and finetune it and the second one is to use the frozen network as a feature extractor and use those features to train a smaller network (see Figure 3). SSL training of the model is generally recognized as pretraining as another round of training or finetuning is expected later. The second stage of SSL only requires a very small number of annotated data as the network is already familiar with the data distribution. In our work, we use self-supervisory signals to pretrain the object recognition model.

SSL techniques are being developed rapidly for benchmarking tasks. Many novel techniques have been introduced by the research community (Ho et al., 2020; Jakab et al.,

2020; Kundu et al., 2020; Mittal et al., 2020; Zhan et al., 2020). SSL methods are powerful and have shown promising results in low-annotated-data situations. One of the major advantages of SSL according to research (Chen, Kornblith, Norouzi, et al., 2020; Chen, Kornblith, Swersky, et al., 2020) is that powerful representations can be learned with smaller models. However, it is vital to investigate the usage of SSL with applications in different domains (Kolesnikov et al., 2019b). Existing works (Dwibedi et al., 2017; Georgakis et al., 2017; Khalil et al., 2013) have used cropped-object placement in random backgrounds as a method of generating a synthetic training signal for object detection and localization. Following these, we also used a similar SSL technique to pretrain our reflection localization network. We cropped the corneal area from annotated images and placed them randomly on generated backgrounds (Figure 4).

Apart from these tasks there is a separate set of tasks called contrastive pretext tasks. Recent work has shown the superiority of SSL models that are pretrained with contrastive signals (Chen, Kornblith, Norouzi, et al., 2020; He et al., 2019; Hénaff et al., 2019; Misra & Maaten, 2020; Oord et al., 2018). Contrastive learning targets to use automatically generated contrastive supervisory signals. In our work, we used a contrastive SSL technique called SimCLR, which is presented by Chen, Kornblith, Norouzi, and Hinton. SimCLR

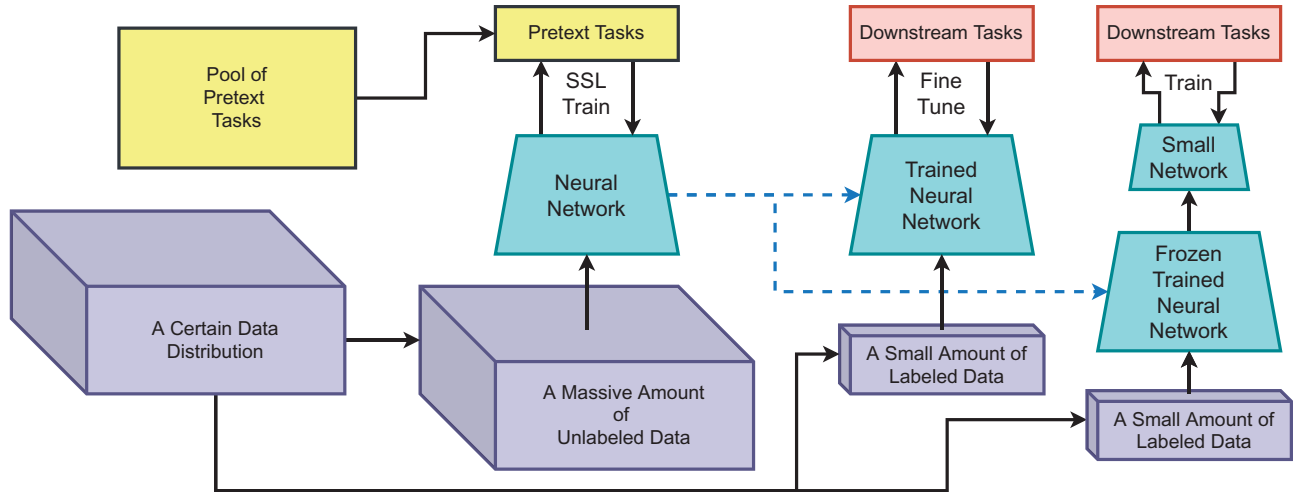


Figure 3. Overview of Self-Supervised Learning.

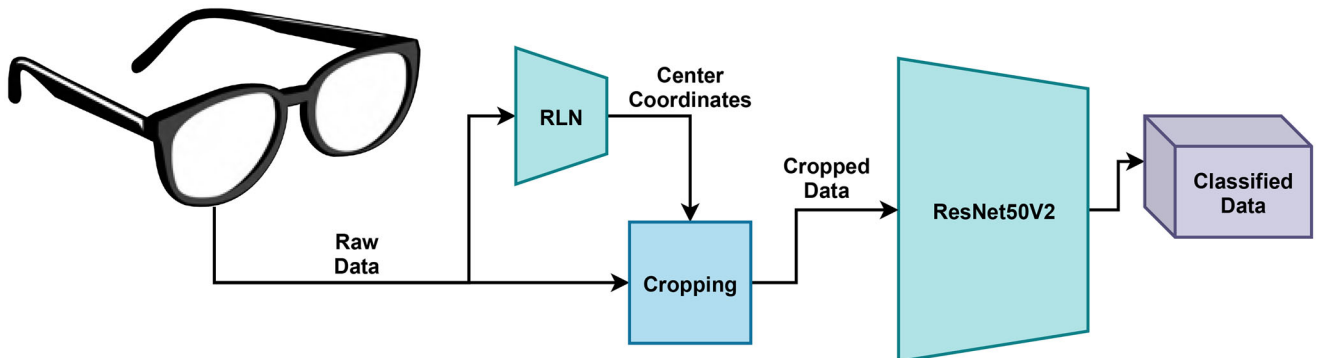


Figure 4. System in Inference Mode: Prototype device records images at 15 frames per second. The Reflection Localization Network (RLN), a light weight neural network, detects the center of the pupil. Another script then crops the reflective area of the eye using the coordinates given by RLN. Finally, a ResNet50V2 network classifies the cropped images.

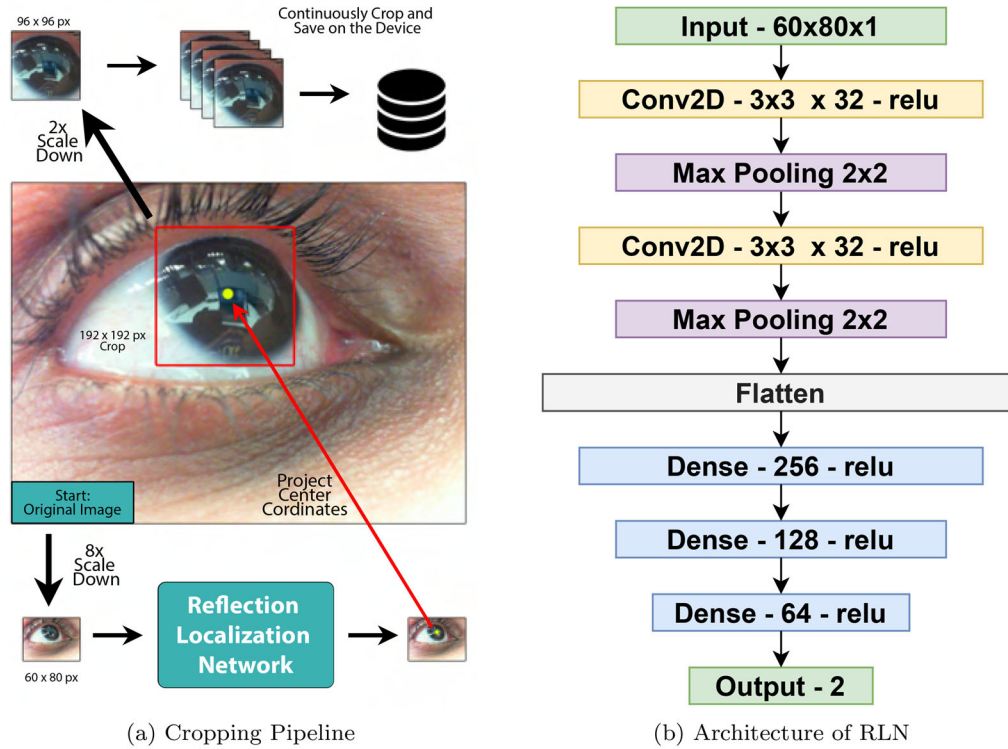


Figure 5. Figure (a) shows the pipeline of cropping. Figure (b) shows the Reflection Localization Network (RLN) used for localizing the center of the reflective area. This network was pretrained with more than 50 million generated images before finetuning with labeled examples. (a) Cropping Pipeline. (b) Architecture of RLN.

seeks to determine if two augmentations are stemming from a single image or different images. SimCLR uses several data augmentation methods such as random cropping and zooming, color alterations, noise, and blur. A program automatically make two augmentations of the same image by randomly selecting two augmentations. Then the network tries to make two augmentations from the same image closer while making two augmentations of different images farther. The intuition behind using this technique is due to the nature of our data. Reflections in eyes are contained to a specific region of the image. Therefore, trying to learn if two augmentations are from the same image should make the network focus on that specific region with reflections, which is the image's dynamic region, irrespective of lighting conditions, eye color, etc.

3. System overview

Our system consists of a hardware component to capture the corneal surface reflections from the eyes and a software component to process the captured images to recognize the focused object in the reflection.

3.1. Hardware

For the hardware, we used a 3D printed eye tracker from Pupil Labs⁴ together with an enclosure-removed Microsoft HD-6000 camera⁵ (see Figure 1). The camera can capture 480×640 pixels images at 15 frames per second. This single-camera mounted to the frame is worn as a headset to

capture corneal surface reflections. An example image captured by this camera is shown in Figure 5(a).

3.2. Software

The software component of the system consists of three stages. First stage detects the center of the reflective area and the second stage crops the reflective area from the image. The third stage determines the focused object in the cropped image. Following sections provide an overview of each stage.

3.2.1. Reflection Localization Network (RLN)

The first stage is a custom-made Convolutional Neural Network (CNN) trained to detect the center of the reflective area of the eye. We call this network, the Reflection Localization Network (RLN). Inputs to the network are 60×80 pixels grayscale images. RLN consists of two convolutional blocks and a single fully-connected block as illustrated in Figure 5(b). Both convolutional blocks of the RLN follow the same architecture with 32 convolutional filters of two dimensional 3×3 kernels followed by a 2×2 two dimensional max pooling layer. Both convolutional and max pooling layers have ReLU activation. The output of the second convolutional block, which is the max pooling layer, is flattened before connecting to the fully connected (dense) block. The fully-connected or the dense block has 3 dense layers reducing the number of parameters at each layer. The three dense layers have 256, 128, and 64 parameters respectively. All the dense layers have ReLU activation. Finally an

output layer of 2 parameters determines the x and y coordinates of the center of the reflective area. RLN has 1.9 million parameters in total.

3.2.2. Cropping the reflective area

The second stage of the software pipeline crops the reflective area of the input images using the center coordinates provided by the RLN. A custom python script carries out this task. First, the original RGB images of 480×640 pixels are scaled down to 60×80 pixels. Then, the scaled images are converted in to grayscale to fed into the RLN to detect the reflection area's center. The normalized floating-point center coordinates (between 0 and 1) predicted by the network are multiplied by 480 and 640, respectively, to project to the original size. A box of 192×192 pixels is then cropped from each image centering the projected coordinates and scaled down to 96×96 pixels. Figure 5(a) illustrates this procedure.

3.2.3. Classification

The third stage is a CNN of ResNet50 Version 2 (ResNet50V2) (He et al., 2016) architecture that classifies the objects using the cropped and re-scaled images. ResNet50V2 is a very popular and more complex CNN compared to RLN. ResNet50V2 is larger in size and has more features in the architecture. It consists of 5 convolutional blocks and a fully-connected block. Residual connections are added between convolutional blocks and a dropout layer is added in the fully connected block for regularization. The entire network has over 25 million parameters in total. Number of parameters in the output layer is changed according to the number of classes being classified. A softmax activation function is used in the output layer for classification. Figure 4 illustrates the entire data capturing and processing pipeline of our system.

4. Model development

4.1. Data collection

We collected our own dataset for model training and evaluation as there was no corneal surface reflection dataset available. Two data collection phases were carried out to collect a large amount of unlabeled data for SSL training and a small amount of labeled data for the supervised finetuning.

4.1.1. Unlabeled data collection in-the-wild

4.1.1.1. Participants. 17 people (6 females and 11 males; age $M = 26.9$, $SD = 4.5$) from the university community were recruited for this in-the-wild data collection. No personal characteristics were filtered in the participant selection process and the only condition was to have normal vision. This was made clear in the participant information sheets and all the participants signed the consent forms before the data collection started.

4.1.1.2. Apparatus. We used the web camera mounted to an eye-tracker frame as described in Section 3 for this data

collection session. The web camera supports many prominent operating systems such as Windows, Mac OS, and Ubuntu. This enables the participants to use their own computers for data recording.

4.1.1.3. Procedure. The prototype was handed over to each participant. We demonstrated how the video recording can be carried out using the default camera application of their operating systems. They were told to wear the device at any convenient environment, such as home or work. We encouraged them to record data in versatile settings rather than in monotonous settings. Also, the participants were given the flexibility to record in multiple sessions scattered over two days. They were expected to record a total of approximately two hours worth of videos. Follow up discussions with the participants informed us that data had been recorded in different scenarios such as while watching TV, cooking, working, and having casual conversations.

Conversion of all the video segments to frames generated a dataset of 2.1 million images (of size 480×640 pixels) within 6 weeks. This dataset was used for SSL-Pretraining of the object classification network.

4.1.2. Labeled data collection for object classification

Supervised finetuning for the downstream task requires at least a small labeled dataset. One way to obtain such a dataset is by labeling some of the images from the already collected unlabeled dataset. This task poses two main challenges. First challenge is the inability of human annotators to annotate with 100% confidence given the tiny representations of the reflected objects and the ambiguity of the focused object. Second challenge is having to go through the entire 2.1 million dataset to find equal amount of data points for each class in this multi-class classification settings. Most importantly, even going through the entire unlabeled dataset does not guarantee to have images from all the classes as the data collection was in-the-wild. Therefore, a second data collection session was conducted to collect a small labeled dataset. The following ten classes were selected for the labeled dataset based on objects found in typical office environments.

1. Desktop screens containing documents (PDFs, web articles, spread sheets, etc).
2. Desktop screens containing media (movies, YouTube, photos, games, etc).
3. Laptop screens containing documents.
4. Laptop screens containing media.
5. Mobile phone screens containing documents.
6. Mobile phone screens containing media.
7. Real documents (books, printed material, files, etc).
8. People (in various postures such as sitting on chair or couch, standing, leaning, etc)
9. Blink (closed eyes).
10. Other (Not specifically looking at something in close vicinity).

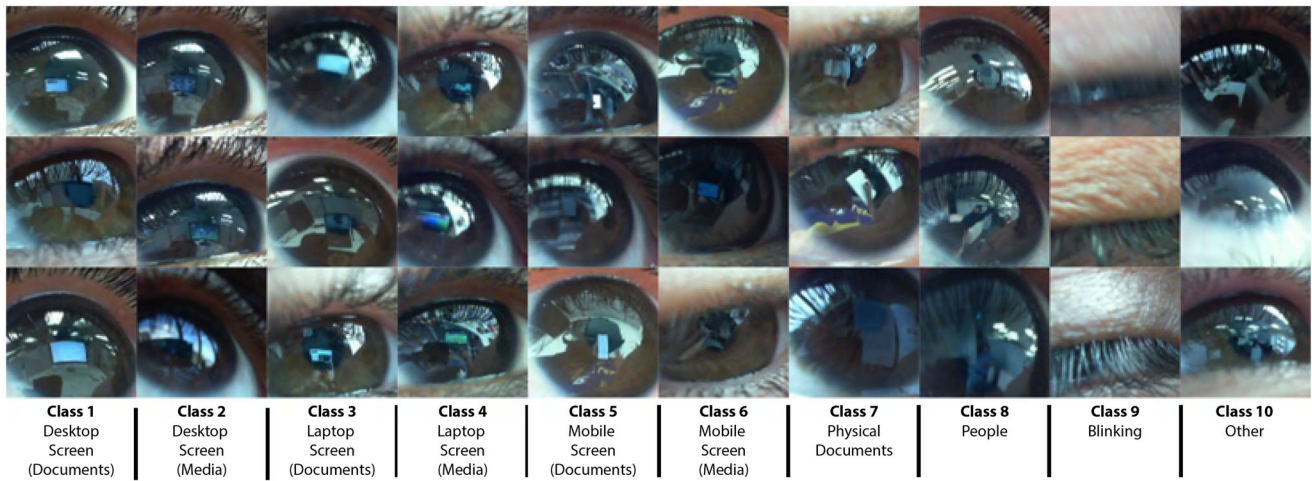


Figure 6. The figure shows 3 examples for each class for the 10 classes we collected. Images shown here have been automatically cropped by the trained RLN.

Blinking was added as a class to have additional information for work related lifelogging. For instance, blink rate has been shown to have correlations (Maffei & Angrilli, 2018) with stress, focus, and mental demand. Therefore having blinks recorded along with the object reflections will allow the lifelogging system to provide insights about these psychological parameters. “Other” class was selected as negative training examples to add contrastive learning to the model training. Images cropped by RLN while the person is looking into extreme angles or while moving the eye really fast can be indeterministic as they do not clearly show the reflection. In these cases, the image was also tagged as “other” so that the model could learn to classify those instances as “other.”

4.1.2.1. Participants. We recruited 9 participants (2 females and 7 males; age $M=27.3$, $SD=3.6$) from the university community. They all had normal eye-sight. Similar to the previous data collection, participants signed consent forms before the data collection started.

4.1.2.2. Apparatus. The same apparatus described in Section 4.1.1 was used for the labeled data collection session. For the multi-user evaluation purpose in our work, we selected ten office related object classes such as documents, screens, and people for lifelogging in work environments.

4.1.2.3. Procedure. Data collection was carried out in the participants own work-spaces at the university. Two researchers participated in the data collection session. During the data collection, each participant was given the control to click a button to record one image. Instructions were to look at the required object or the person, in the normal way they would do it, and press the button when not blinking (except for blinking class). This ensured that the label is in fact the object (ground truth) that the person was focusing. Lighting settings or any other environmental parameter was not controlled during the data collection, and the only hard restrictions were to look directly at the object. 15 data points were collected for each class from each participant which resulted in 1350 images. Some examples from

the collected dataset for each object class are illustrated in Figure 6 (the images displayed in the figure are cropped by our Localization CNN).

4.1.3. Labeled data collection for RLN

A supervised dataset is required to train the localization CNN to detect the center of the pupil. However, having access to the unlabeled and labeled datasets collected for object classification removed the necessity to conduct another separate data collection as existing data can be annotated for localization. We annotated randomly selected 2000 images with the center (x,y) coordinates of the pupil using a custom python script, which is also included in our opensource code-base. The script assists the annotator with a circular contour that can be aligned with the pupil to easily annotate the center thus significantly reducing human error.

4.2. Reflection Localization Network (RLN)

Since there was no existing solution to detect the pupil from close proximity RGB images containing reflections, we developed our own model (RLN). As described in Section 4.1.3, we randomly selected 2000 eye images from the large image pool we collected and annotated them with (x,y) coordinates of the center of the ROI or the reflective area. Input image size significantly affects the neural network size when using a CNN with fully connected layers at the end. Therefore we scaled down the images eight times, resulting in frames of 60×80 pixels. That was the smallest we could go without visibly pixelating the image. The idea here is to scale down the image, detect the center of the reflection, project the coordinates to match the original size, and crop (see Figure 5(a)). However, to achieve this, training a blank network with 2000 images was not possible due to the overfitting problem. Therefore, we used a synthetic pretraining-based technique following prior works (Dwibedi et al., 2017; Georgakis et al., 2017; Khalil et al., 2013) that use random placing of objects in dynamic backgrounds to generate synthetic data. All the images used for RLN are grayscale as RGB information is not required to crop the pupil area.

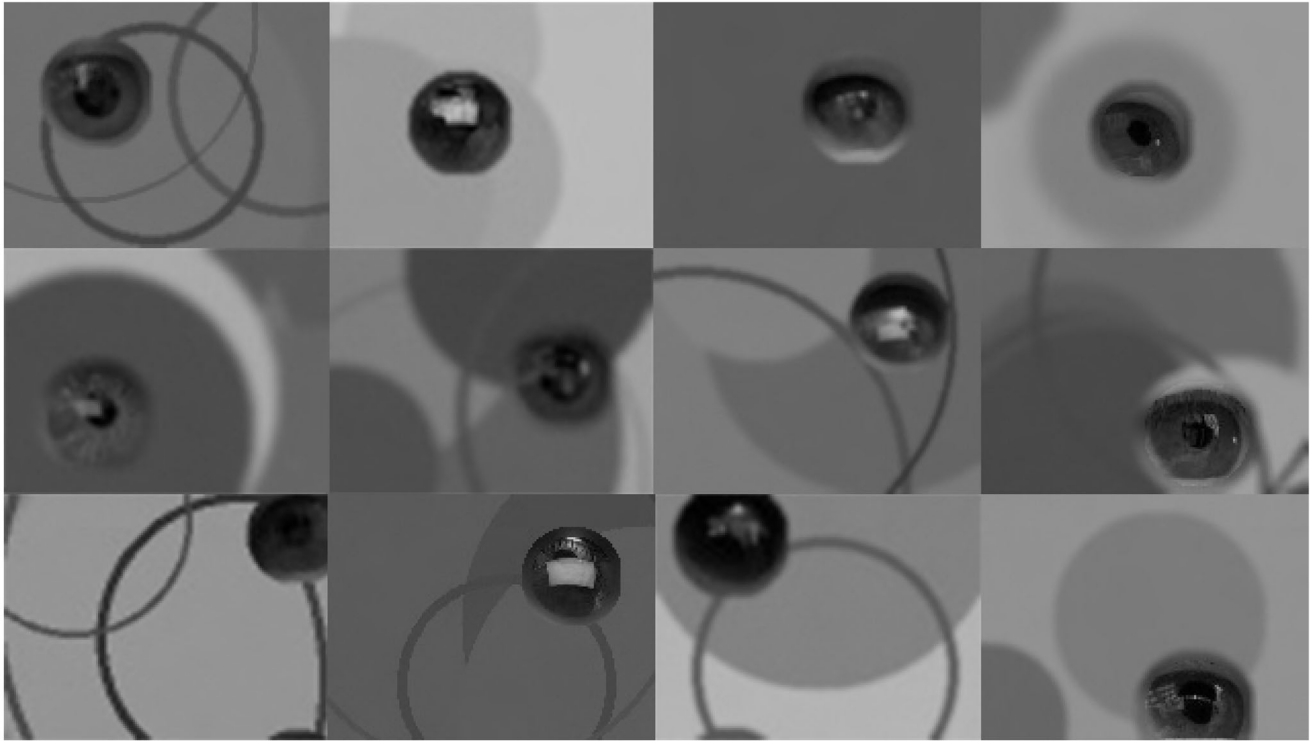


Figure 7. Examples for generated synthetic images that are used to pretrain the RLN.

4.2.1. Synthetic pretraining

Prior works Law and Deng (2022); Yang et al. (2021) have explored using synthetic backgrounds for pertaining object detection networks. However, this concept has not been applied in pupil localization models in corneal surface reflection analysis. In addition, we use a different technique to create synthetic backgrounds using circular shapes.

We created a frame generator which places real corneas extracted from the labeled data on generated backgrounds. The generator starts with a randomly bright plain background. Then a random number of circular shapes with random fill colors are placed on the background. Then circular lines of random sizes and brightness are placed on top. The reason for selecting circular shapes is to make the task a bit hard for the neural network since the cornea, eyelids, and eyelashes are also circular. Finally, a Gaussian blur with a random strength was applied to all the background randomly, including a no-blur option. After generating the random background, an image is randomly selected from the labeled dataset of 2000 images to extract the cornea. The cornea was circularly cropped from the image around the (x,y) coordinate using a random radius varying between a small margin. This was done to ensure that the model is not trained to detect a fixed size cornea. After cropping the cornea, an edge-blur of random size and strength, including a no-blur option, was added to the cropped circular shape to blend the cornea with the generated background. Finally, the cropped and processed cornea was placed at a random place determined by a random coordinate which is tagged to the image as the center label. Some examples of the generated images are shown in Figure 7.

4.2.2. Hyperparameter tuning

Following the development of the synthetic image generator, we developed and experimented with several basic CNN architectures for center detection. We evaluated 15 different neural networks with 2,3,4,6 and 9 convolutions layers and pooling layers structured in different blocks followed by several fully connected layers. For instance, one structure had two blocks of three convolutional layers and a max pooling layer while another had three blocks of two convolutional layers and a max pooling layer. We also changed the number of dense parameters and kept the model size below 3 million parameters all the time to make sure we do not exceed the size of already available models. The main idea was to find the smallest network with the best performance. The synthetic image generator is connected as the input the model to generate images while the training. Each model was trained with generated images for 100 epochs where each epoch was 1000 steps of batch size 512. Since we could generate infinitely different examples due to the random nature of the backgrounds, we did not intentionally repeat training examples for two epochs, and just let the generator provide random inputs for each step. Each network architecture we experimented with saw more than 50 million ($100 \times 1000 \times 512$) generated examples in total. Given the small network and input sizes, training a single network only a few hours on an Alienware R13 2017 laptop with a 6GB NVIDIA GeForce GTX 1060 Mobile GPU. We selected the network that gave the best results (validation Mean Squared Error value = 0.01). The final selected an architecture of 1.5 million parameters which had two convolutional layers, each followed by a max-pooling layer, and three dense layers at

the end of the network (shown in Figure 5(b)). We call this pretraining since this comes before supervised training. The selected model's pretrain validation MSE (Mean Squared Error) was reduced to 0.01 for the regression loss. Coordinates were given as floating points between 0 and 1.

4.2.3. Supervised finetuning

We used the 2000 annotated images with the center coordinates of the reflection area. Since the coordinates could be easily altered by carrying out some augmentations, we did horizontal and vertical image offsetting by a random amount and flipping left-right. The program made sure that the cornea is not cut off by the offset. Before doing offsets and flips, we normalized all the images (50 to 180 – 8 bit) for exposure since participants with different skin tones recorded the frames in different lighting conditions. We filled the gaps in offset images with a medium-grey (120 – 8 bit) color to match participants' average skin tone in normalized grayscale images. To balance the ratio of real examples to augmented examples, we only created 6000 augmented examples. Therefore, the total amount of labeled data was 8000. We randomly selected 6000 images for training, 1000 for validation, and 1000 for testing. We finetuned the network for 20 epochs with the 6000 examples and achieved a test MSE of 0.0013. Considering a 60×80 image, this error value corresponds to 2.5 pixels, which is acceptable for detecting the center. The model performed well in capturing the reflection area (to illustrate, all the images in Figures 2 and 6 are cropped using this network).

4.2.4. Comparison with existing models

The intuition behind the development of RLN for localization was to have a smaller network with good performance. However, to evaluate if our custom model performs closely with the existing models, we conducted an evaluation with two other models: ResNet50V2 (He et al., 2016) and MobileNet (Howard et al., 2017). The same dataset we annotated for the supervised finetuning of the RLN was used for the experiment and a regression layer of size 2 was added at the end of each model. Weights for the other two models were initialized from ImageNet (Russakovsky et al., 2015) checkpoints. Training was carried out for 40 epochs and the best validation accuracy from all 40 epochs are reported.

Results clearly show that the biggest model we evaluated, the ResNet50V2 (He et al., 2016), provided the best results. RLN has a smaller Mean Square Error (MSE) than the MobileNet (Howard et al., 2017) that has 1.3 million more parameters than ours. RLN provides satisfactory performance with the a smaller amount of parameters, which will enable fast and low-power operation in embedded systems in the future.

Table 1. Comparison of our custom CNN and existing popular models.

Model Name	Number of Parameters	Validation MSE (lower is better)
ResNet50V2	23.6 million	0.0008
MobileNet	3.2 million	0.0165
Ours (RLN)	1.9 million	0.0013

4.3. Object recognition in eye reflections

With our RLN network and a custom script, all the 2.1 million unlabeled frames we collected were cropped and re-scaled to 96×96 pixels, which we selected based on the SimCLR (Chen, Kornblith, Swersky, et al., 2020) work (see Section 3).

Using SimCLR for our work is motivated by two main factors beyond the well established robustness of SSL against the scarcity of labeled data. All our eye images looked similar, except what is reflected on the cornea. Therefore, we assumed that a contrastive loss that tries to maximize cosine similarity between two augmented outputs from the same image would learn to focus on the volatile area of the image, which is the reflection. Secondly, the authors of SimCLR have also shown that the performance of bigger traditional models can be achieved by smaller models trained with SimCLR framework. Therefore we believe that using SSL will enable lightweight models that are easily adaptable (Kolesnikov et al., 2019a) for corneal surface reflection related applications in the future (Table 1).

4.3.1. Further SSL pretraining with domain specific data

Inspired by the recent findings that highlight how further-pretraining of SSL networks with domain-specific data can improve the results (Gururangan et al., 2020), we pretrained the network further using the the same NT-Xent (Sohn, 2016) loss function (see Equation 1) used by SimCLR. In addition 2.1 million is a comparatively smaller amount for an SSL task, which supports the decision to further pretrain from a well-trained checkpoint as opposed to pretrain from scratch.

$$Loss_{i,j} = -\log \left[\frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{(k \neq i)} \exp(\text{sim}(z_i, z_k)/\tau)} \right]; \quad (1)$$

$$\text{where, } \text{sim}(u, v) = \frac{u^T v}{\|u\| \|v\|}$$

We used the pretrained SimCLR checkpoint⁶ provided by the authors with ImageNet as the base SSL model. We kept all the parameters constant as the SimCLR article, including the input size of $96 \times 96 \times 3$ and the batch size of 512. We pretrained a ResNet50 architecture using Pytorch SimCLR implementation on a Tesla v100 32GB GPU. We trained it for 80 epochs that took about one month. We randomly separated 100, 000 examples for validation.

4.3.2. Supervised finetuning

In total, we had 1350 labeled examples from 9 participants. We rotated the participants to have a leave-one-out 9-fold cross-validation (keeping one participant for testing and 8 for training for each instance) where the network was

finetuned 36 times for all the class variations. No hyperparameter was changed during the cross-validation evaluation eliminating the need to have a separate validation set. A dense layer of size equal to the number of classes was added on top for each class variation. For each instance, we finetuned the entire network for 20 epochs with a small learning rate of 0.00005. We used a batch size of 16, as we had a limited amount of labeled data. Results of the classification for different class combinations are shown in Section 5.1.

5. Evaluation

5.1. Classification

5.1.1. Accuracy & F1 scores

Average 9-fold cross validation accuracy for 10 classes was 57.50% as shown in Table 2. Further analysis of these 10 classes revealed that some classes had similar or unclear examples, thus making confusions as illustrated in Figure 8(a). Due to the confusions among the screen classes and a higher confusion of “people” with “other.” Therefore, we conducted more evaluations by combining those confusing classes, but still maintaining main classes for practical application scenarios. For instance, first, we combined all computer screens into a single class and all mobile phone screens to another class. This resulted in 6 classes.

Figure 8(b) shows the confusion matrix for the 6-class variation. The 6 class variation improved the average accuracy from 57.7% to 71.5% and the F1 score from 52.7% to 67.9%. Following the same approach, later we combined all screen classes to a single class creating 5 classes (see Figure 8(c)) in total. As the last step, we combined people and other classes together to get a 4 class variation. Accuracies and F1 scores—a metric indicating the balance between precision and recall—of for 10, 6, 5, and 4 classes are presented in Table 2 as percentages.

5.1.2. Effect of blink class

As mentioned in Section 4.1.2, blinking is separately detected as a class given the insights it provides for

lifelogging applications. However, as seen from Figure 8, detecting blinking is comparatively easier task than the other classes. Table 3 presents average accuracy excluding blink for each class variation to illustrate the accuracy change.

5.1.3. Effect of class combinations

There are two ways to combine classes. The trivial one is to combine the classes after predictions. However, this is inefficient since this does not reduce the number of parameters of the network according to the number of classes. The final layer of the network can be reduced to have number of logits equal to the number of classes as the prediction layer uses Softmax. Therefore, we trained the model again for each variation by changing the last layer each time.

Following prior work (Withana et al., 2015), we conducted a one-way ANOVA analysis to see if the accuracy increases for lower number of classes are significant. Based on one-way ANOVA, there was a significant effect of class variations ($F_{3,35} = 13.01$, $p < .05$). A post hoc analysis using Tukey’s HSD revealed that the accuracy of the ten class variation was significantly lower than all other variations (6, 5, and 4 classes). There was no significant difference between 6-classes and 5-classes, 6-classes and 4-classes, and 5-classes and 4-classes variations. This mean that reducing the number of classes from 6 to 5 or 4 does not significantly increase the accuracy (see Figure 9(a)).

When some classes were combined, we kept 15 examples per participant per class by randomly selecting uniformly distributed examples from combining classes. Therefore the total numbers of training examples were 1200, 720, 600, and 480. However, even with a case of lowest annotated examples (i.e 480 labeled samples), the results show that our training did not overfit the model to the data as the average cross-validation results are increasing when the number of classes is dropping. As seen by Table 2, F1 scores of all the participants tags along well with the increasing accuracy values and it shows that, for all the participants, false positives, false negatives, true positives, and true negatives are balanced. This shows that the SSL training is stable and still robust in low data situations.

Table 2. Class Variations—6 class variation was a result of combining all desktop and laptop screens to a single class and all mobile screens to another classes.

P	10 Classes		6 Classes		5 Classes		4 Classes	
	Acc	F1 scr	Acc	F1 scr	Acc	F1 scr	Acc	F1 scr
1	72.67	66.47	84.44	84.03	86.49	85.78	88.33	88.32
2	52.67	48.99	71.11	67.4	66.22	63.38	76.67	69.76
3	53.33	50.64	74.44	69.21	87.84	87.86	86.67	86.61
4	62.00	59.5	70.00	69.48	86.49	85.77	81.66	81.38
5	41.33	32.13	63.33	55.25	60.81	57.36	75.00	66.67
6	66.00	59.88	64.44	57.99	71.62	69.72	70.00	61.04
7	54.00	51.23	65.56	60.91	67.57	62.07	83.33	81.52
8	52.00	46.28	68.89	65.53	77.02	76.41	88.33	88.02
9	65.33	58.92	81.11	81.02	75.68	74.02	83.33	83.57
Avg.	57.70	52.67	71.48	67.87	75.53	73.60	81.48	78.54

5 class variation was a result of combining all screens to a single class. 4 class variation consisted of screens, documents, other, and blink. In all these scenarios, we maintained 15 examples per class per participant to maintain class balance. This table present 9-fold cross-validation (leave-one-participant-out) results (Acc(%) and F1 score) and the overall averages in the last row (bold). P = Participant No.

5.1.4. Ablation studies

We selected the 10-class classification for ablation studies as it was the most demanding task for the network. For ablation studies, we split the labeled dataset into train, validation, and test sets. We randomly selected 5 participants for the training set, 2 participants for validation, and 2 participants for the test. First experiment was to use classical transfer learning by taking a ResNet50V2 checkpoint trained for object recognition with ImageNet (Russakovsky et al., 2015). The entire network was finetuned with our labeled dataset. As seen from Table 4, it also performs on par with the SSL domain-specific pretrained model but slightly lower in F1 score. As seen from the second row of Table 4, the SSL model without further pretraining on domain-specific data performed significantly poorly than the other two variations.

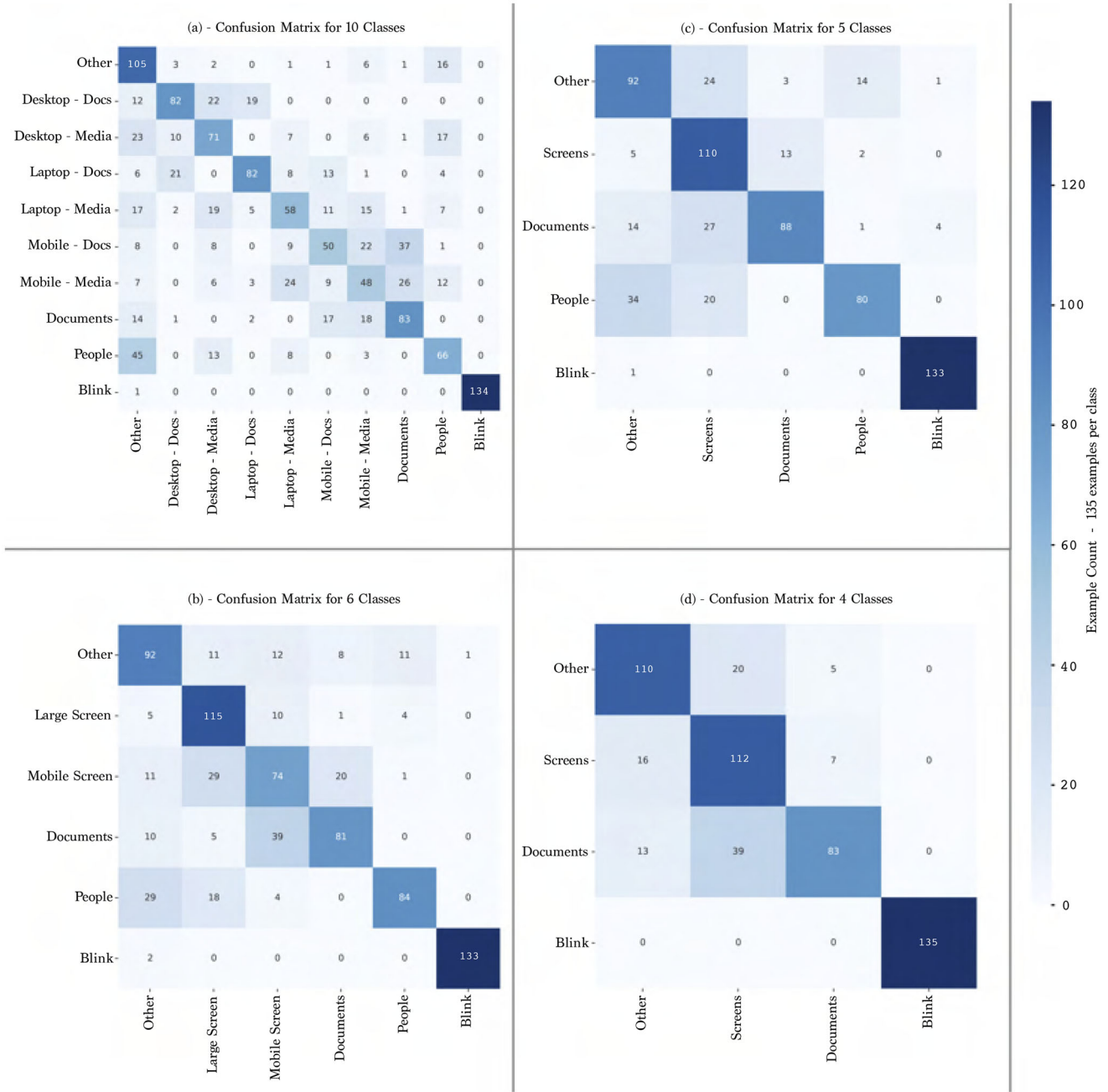


Figure 8. Figure shows the confusion matrices for multiple class variations.

Table 3. Average accuracy (%) of class variations concerning all the participants—excluding blink class.

	9 Classes	5 Classes	4 Classes	3 Classes
Average accuracy without blink class	53.09	65.48	68.51	75.31

This ablation study shows some critical points. First, it shows that further-pretraining SSL networks with domain-specific data improves the performance. In our application, it improved the performance considerably, and this can be useful for other computer vision applications in other domains as well. The next important point observed from Table 4 is, our method performs only slightly better than transfer learning on ImageNet. However, as shown in prior work (Chen, Kornblith, Swersky, et al., 2020; Hendrycks

et al., 2019; Kolesnikov et al., 2019a), our SSL approach has the potential to be used for challenging situations where a high-quality representation has to be learned from data with fewer labels or uncertain labels.

6. Feasibility for lifelogging application development

Having evaluated the model, we conducted an in-the-wild study to investigate the applicability of our system for lifelogging. Intention of this study was to see the performance of our system in-the-wild and also get feedback from users regarding the insights that it can provide. Evaluating our system with real human feedback to understand the human context for the application development is vital for the

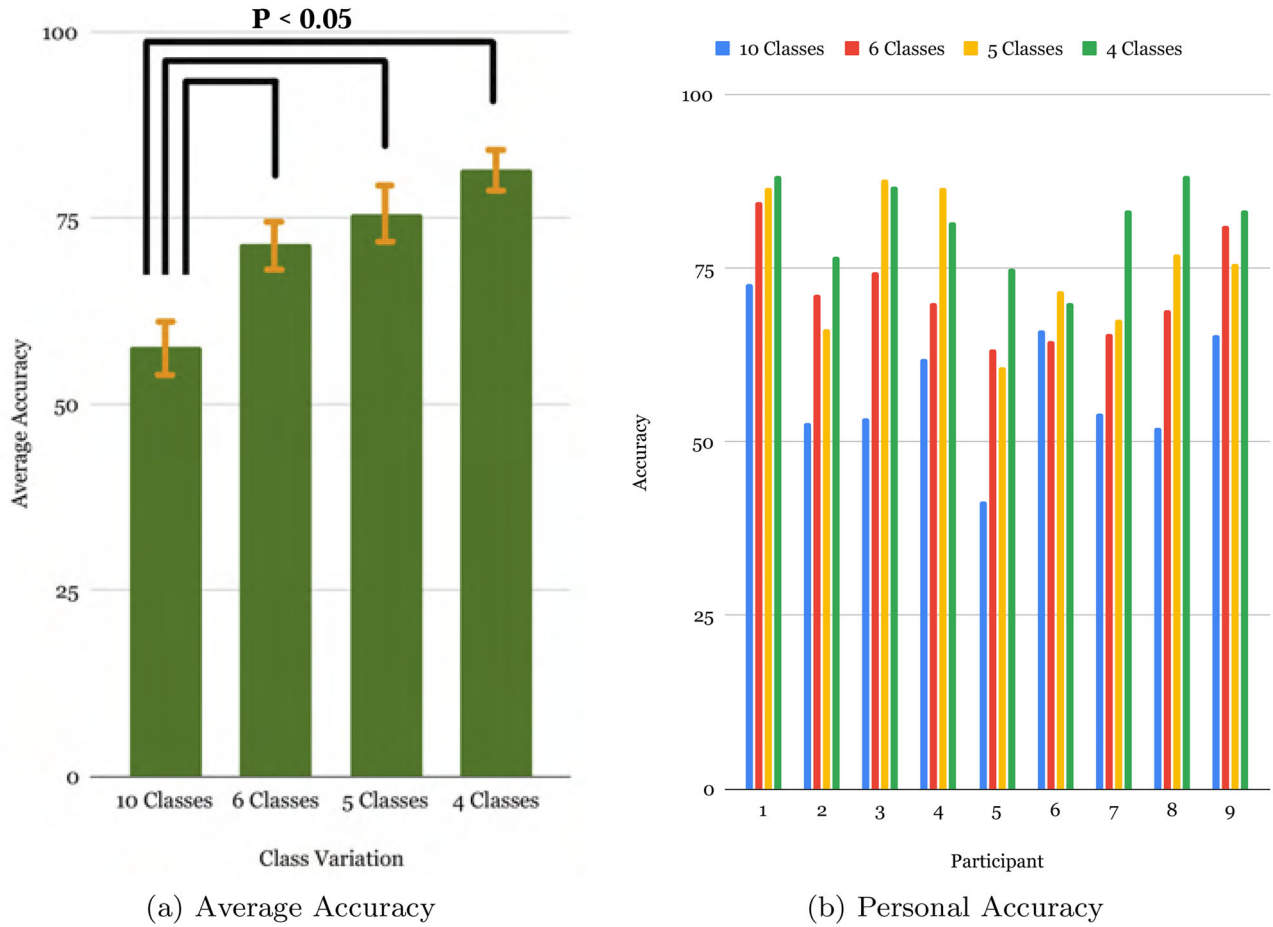


Figure 9. Figure (a): Accuracy values for each class variation. 95% confidence interval is illustrated in yellow bars bar. 95% significant differences among the class variations are drawn with black lines. Figure (b): Accuracy values for each participant clustered by class variation. (a) Average Accuracy. (b) Personal Accuracy.

Table 4. Ablation Studies. Different ways of using the ResNet50V2.

ResNet 50 Training Variation	Accuracy	F1 Score
Non-SSL: ImageNet transfer learning	71	70.62
SSL: SimCLR ImageNet pretrained + without pretraining on eye images	41.67	40.72
SSL: SimCLR ImageNet pretrained + with pretraining on eye images	71.67	72.72

Human-Centered Development of the system T. Kaluarachchi et al. (2021); Margetis et al. (2021); Shneiderman (2022a, 2022b).

6.1. User study

6.1.1. Participants

We recruited 6 participants (age $M = 27.6$, $SD = 2.3$) from the university community. They all had normal eye-sight. Similar to the previous data collection, participants signed consent forms before the data collection started.

6.1.2. Apparatus

We used the same apparatus in the previous in-the-wild data collection session (see Section 4.1.1). We used the 5-class model we trained for this study as it performs better than the 10 classes and provides useful information about the main object categories we had in the 10 classes.

6.1.3. Procedure

We handed over the device to each participant and asked them to collect videos during the day. We asked the participants to keep a note on what they did during the video recordings. The specific task was to note down the objects—out of screens, documents or books, people, and other—they were looking at. Each participant sent us video files that were 10–15 minutes long.

6.1.4. Data analysis

We converted all the video files to frames (around 75,000 frames in total) and each frame was linked to the participant number. We cropped all the frames using the RLN pipeline and then classified all the frames using the 5-class ResNet we trained. Since the activities of the participants were not controlled in any manner, classes were imbalanced according to the results of the model. About 54% of the frames

Table 5. 30-Frame Accuracy (%) of the Preliminary In-the-wild Evaluation.

	Screens	People	Docs	Blink	Other
30-Frame Accuracy	76.7	63.3	56.7	90	66.6

were classified as screens, 1% people, 12% blink, 5% documents, and 28% other.

We randomly selected 30 from each class to a total of 150 frames for all 5 classes. Then we separated the 150 frames by participant. We followed-up with the participants to help us annotate their frames. They watched the video with us and referred to the notes to assist us in this annotation. We took these annotations as the ground truth for our quantitative performance analysis. In addition, we conducted open ended discussions to get their thoughts on the performance of our model.

6.2. Results & implications for application development

6.2.1. Acceptable starting performance

Table 5 shows the class-wise accuracies comparing the ground truth with the classified class. The average accuracy for all the classes is 71.7% with the blink class and 67.2% without the blink class. These are on par with the values 75.5% and 68.5% from Tables 2 and 3. These results show that the system performs consistently in an in-the-wild situation. Without any further training or modification, our model can report blinks with more than 90% accuracy. Blink rate is shown to have indications regarding attention and task difficulty (Maffei & Angrilli, 2018) which can be used as an added parameter for lifelogging. Current performance is a good starting point, as participants pointed out that currently they do not have a way to automatically get the information that our system provides. However, the model accuracy can be improved when more people start using by leveraging human-in-the-loop deployment (Kaluarachchi et al., 2021).

6.2.2. Creating awareness is useful

Three participants found this interesting as their initial impression. P1 stated, “It sounds interesting to me; I have never seen such a capability before.” This also informs us about the potential of integrating into a consumer product. P6 said that he didn’t want to know his heart rate (HR) before; but after his smartwatch constantly provided the HR and reports, he had started checking it often and has now become a useful feature. P6 mentioned that same concept may apply to our system after seeing screen times and people-times over a period of time. Overall, all the participants had a positive attitude towards using a system that can provide reports of what the user was looking at during the day. P3 stated, “I like to reflect on what I do on a day and therefore, I’d use this daily if it’s not a hassle to use it.”

6.2.3. Presentation of results need attention

Different participants wanted different presentation strategies for the results. P1 wanted to see the different classes in a pie chart to see at a glance, with the option of seeing the

timeline of activities if needed. P5 also wanted a similar presentation. P2 and P4 wanted to see stress or mental demand related indicators to be highlighted more in the report. P6 stated, “I want insights, not pure values. I usually do not care how long I spend in front of screens or read a book. I want to know actions. For instance, my watch tells me you have been sitting for a long time. Now go for a walk. Something like this. You spent too much time with screens today, so don’t work tomorrow.” P3 also wanted to see continuous chunks of screen time rather than having daily percentages. Overall, the discussions showed that the presentation of results need to be designed carefully allowing for individual user preferences.

6.2.4. Importance of features are Person-Dependent

Similar to result presentation, importance of the features are personal. P1 and P5 liked having the screen times for health concerns. P5 even wears a screen light filtering glass. P1 wanted to control the screen time and would like to get reports for a long period of time to try several screen time controlling strategies and get feedback. P2 and P4 saw the potential of getting stress and mental demand indicators as the most important feature. P2 wanted to see screen time related stress and P4 wanted to recognize patterns of mental demand and stress so that those time slots can be used effectively for demanding tasks. Overall, the least preferred feature was the document or book time, but this is personal thus can change for a different set of people.

6.2.5. Hardware improvements are necessary

Three out of the six participants wear spectacles and given a seamless hardware design in the future that is not bulky, those three had no concerns wearing spectacles during the day. However, The other three people had concerns regarding wearing spectacles. This shows that people who do not wear spectacles regularly will resist adopting this as is. However, P3 stated that given a useful feature like blue light filter in the spectacles, wearing it during the working hours would be further motivated. Therefore, hardware needs to be improved in terms of usability and integration with other useful features.

Given the rapid advancements in the field of hardware and technology, there are existing spectacle frames with built in miniature cameras⁷ looking inwards. These technologies will be widely available soon, and our approach of using corneal surface reflections with a single camera looking at the eye would be a practical and seamless solution for lifelogging applications.

7. Discussion

7.1. Social acceptance vs privacy

As pointed out by previous works (El Hafi et al., 2017a; Lander & Krüger, 2018) as well, social acceptance is one of the key reasons for using corneal surface reflections. However, there’s a tendency to confuse social acceptance

with privacy. Privacy refers to the access to personal data and its usage while social acceptance refers to a person's willingness to use the device and this is highly dependent on other people's attitude towards the person wearing the device (Kernaghan, 2016; Profita et al., 2013). A device with a camera pointing outwards could have the same privacy settings as a device with a camera pointing inwards such as not saving, or sharing raw images. However the social acceptance is more related to whether a person is comfortable being near a person who has a camera pointing outwards, capturing all background visuals. This is also common to the person wearing the device as the user knows how uncomfortable other people around him can be. Social acceptance is a serious concern as one of the reason for the downfall of even popular commercial products such as Google Glass has been poor social acceptance (Kernaghan, 2016) where "accidentally recording or photographing others" was one of the concerns. When using corneal surface reflections, as we can see from the images illustrated in the article, the images are not much informative other than the focused area. With the resolution we used and due to the convex nature of the eye focusing only on the attentive area, it is practically impossible to capture any unwanted background incidents. We believe that this is the main advantage regarding the social acceptance when using corneal surface reflections, however with a good and transparent data privacy policy.

7.2. Simplicity vs efficiency

In our approach, just using a single camera and no other sensors makes a minimal and simple system. Such system, we believe, can enable the development of new class of life-logging devices in the near future. However, using a single camera without infrared creates some additional challenges such as reduced visibility of the pupil that made it hard for us to detect the ROI using traditional computer vision techniques. As visible in Figure 2, cornea of some eyes (such as a and b) are strong reflectors while only pupil is strongly reflecting in some eyes (such as d and g). One prior work (El Hafi et al., 2017a) has used a mathematical 3D eye model to tackle this problem while another (Lander & Krüger, 2018) has used a object detection model at once. However, we used a small custom CNN to tackle this which has two advantages over the prior works. First one is that no calibration is required compared to a user-dependent model, and the second one is the smaller model size. A model like YOLO-V3 (Redmon & Farhadi, 2018) has about 60 million parameters although it does both detection and recognition. Therefore running this on-device is really challenging and even if it is run on cloud, the entire image (containing less than 10% of useful information) needs to be uploaded to the cloud. However in our approach, the detection model is separate and has about 1.9 million parameters which makes it easier to run on-device and requires a significantly lower number of calculations. On top of that, after cropping, only 96×96 pixels need to be uploaded compared to 640×480 pixels.

7.3. Applications beyond lifelogging

Although we did not explore in this work, there are many other advantages of having a camera looking at the eye. First one, which is the easiest to realize by building on top of our work is activity recognition. Although an image alone can tell what a person is looking at, a sequence of images processed together using a 3D Convolutional Neural Network (Zhou et al., 2018) or a space-time attention based Vision Transformer (Bertasius et al., 2021). When a person is walking, the objects in the corneal surface reflections should become larger gradually since the person is getting closer. We believe that these kind of visual clues can be used by the neural networks to recognize several activities. We envision such activities to be walking, running, some sports, driving, cooking, eating, etc.

In addition, there are other use cases mainly beneficial for health and well-being. Information gained from eye images can be used to derive eye-tracking information and infer mental demand or stress (Fridman et al., 2018; Ren et al., 2014). Research has shown the possibility of using eye images taken from cameras without using any specific imaging devices to determine several diseases. Those include measuring cholesterol levels (Alhasawi et al., 2018; Nurbani et al., 2019), detecting high intraocular pressure (Aloudat et al., 2019) and early detection of some eye tumors (Dimililer et al., 2016). Apart from these specific diseases, the redness in the eyes can be easily detected by images which can be signs of tiredness, sleep deprivation, and many other serious illnesses that even require to visit a doctor (Frings et al., 2017).

7.4. Limitations

7.4.1. More data would have been better

We were able to collect unlabeled data from 17 people to develop SSL-pretrained model. Again, more data would have been better, however, we could not recruit more people mainly due to COVID-19 related lock-downs. COVID situation hindered conducting smooth data collection sessions. In unlabeled data collection, we dropped the prototype in front of some participants' doorsteps as non-contact delivery and cleaned the device after each use. We maintained a gap of at least one day between two participants. However, since annotated data collection needed to be under our supervision, we could not follow the same process. We collected data from 9 people for annotation. This was sufficient for our work to test if our model could work without having to calibrate for each individual user, show the potential of our approach and gain insights. However, collecting more data from more people might have improved the accuracy and performance.

7.4.2. Eye characteristics matter

We achieved satisfactory results, but, going further was hindered by a few challenges that are visible from the results as well as the data we collected. When the images (b) and (d) of the Figure 2 are compared, it is clearly visible that the

intensity of the reflections are not similar. In fact, the reflections of (b) is clearer than of (d). Reason for this is a personal eye characteristic that we noticed, which is the color of the iris. The darker the iris, the better the reflections. While the entire cornea reflects light equally in (b), pupil area reflects more light than the iris region in (d). However, still the entire cornea somehow reflects light. This effect is clearly visible in Figure 9(b) as the eye (b) in Figure 2 belongs to the participant 1 and the eye (d) belongs to the participant 5. As visible from the chart (Figure 9(b)), participant five has a higher effect on lowering the average accuracy. Although this remains a challenging situation now, we believe that this can be made better in the future by slightly fine tuning the model for very few iterations on personal data for the people with lighter reflections. We discuss some possible ways to do this in the future work section.

7.4.3. Different classes looking similar in reflections

The confusion matrix for ten classes, in Figure 8(a), shows higher amount of confusions occur between “people” and “other” classes. We investigated the data to understand the underlying reasons for this. The first one is that in some of the images that are tagged as “other” contains reflections of people in the vicinity as we recorded “other” images without controlling the environments to create real-world data. This might have triggered the network to make confusions among these two classes. Another reason we noticed is that in some instances when two people were talking to each other, they maintained a certain distance and the reflections of the people in those instances were quite small, even we found it harder to recognize. For example, there were instances that we recorded data when two co-workers were talking to each other sitting in their working table, that was around 2 meters apart. In some instances, the person was comfortably sitting on a couch leaning to the back creating unusual angles (see the middle image of class 8 in Figure 6). We believe that these reasons triggered the higher confusions.

Similarly to this, Figure 8(a) also shows confusions within the screen sub-classes and between the document and mobile screen classes. Confusions among the screen-sub classes were not quite surprising as we expected some confusions. However, the interesting finding was the confusions between document class and the mobile screen classes. We examined some data to figure out possible causes and we believe that the reason is similar appearance in size and color. To clarify, typical A4 sized or smaller documents and books on the table appear the same size as mobile phone held a bit closer to the eye. The confusions among the screen classes are mainly due to the different screen sizes and the different seating distances from the screens. Considering the document and media viewing from the screens, it is visible from the confusion matrix (Figure 8(a)) that document sub classes of the screen classes always show lower confusions than the media sub classes. This is mainly due to media being frequently darker than the documents, that most of the times contain more white space. Darker media creates low intensity reflections since the light emitted from the screens are lesser. Therefore media tends to be

confused more with other classes. Interestingly, this confusion for desktop and laptop screens mainly occur with the people class while for mobile screens, it is the document (physical) class. We discuss some of the possible solutions to mitigate above confusion problems in the future work section.

7.4.4. Low light is challenging

Apart from the challenges mentioned above, there is a hardware-related limitation. Since RGB images are collected using visible light cameras, information contained is not sufficient if recorded in very dark environments. The trivial solution for low-light imaging is using infrared (IR) cameras. However, using infrared cameras will also not work here since infrared cameras cannot capture reflections of the corneal surface. As a potential solution, we augmented the RGB camera by removing the default IR filter. Reflections recorded with this camera was sufficiently visible in indoors with artificial lighting. However, when the indoors get natural light, its infrared radiations overshadow the reflections captured by our IR augmented RGB camera. To overcome this challenge, a custom camera needs to be developed to dynamically control the IR filtering mechanism.

7.5. Future directions

Many possibilities for future work emerge from our work in terms of addressing the challenges and also to build on top of this work. It is ideal to collect more data from people with light eye colors because, as shown by our results, corneal surface reflections of darker color eyes generalize well in terms of multiple people. However, there is also a possibility to deploy a continuous learning scheme where the user is prompted once in a while to answer a few questions regarding what the person is looking at. This can be used to fine tune the network to the person on the cloud with the time. In fact, this technique is considered to be a Human-Centered Machine Learning (HCML) (T. Kaluarachchi et al., 2021) approach used in consumer AI products.⁸ Almost all AI systems are improved continuously via secure data collection mechanisms and we believe this will improve the performance of the system over time.

To reduce the confusions among classes such as screens of different sizes, screens with documents and people with other, collecting more data is the best method. However, when this is not possible there are some other possible ways to mitigate this. One way to reduce false positives is to set a probability threshold at the Softmax layer, to categorize an object as “other” based on the confidence. This will not improve the overall performance of the system but will reduce errors. The other way to actually improve the performance is using a similar continuous learning scheme, where the confusions are clarified through prompted questions from the users.

In terms of improving and developing the work further, it is important to build a high fidelity prototype with a usable form factor that can carry out the functions to

validate the usability of the device. This will help evaluate performance in real-world deployment via longitudinal user studies. This will also enable evaluations of power consumption, data usage (which we have estimated to be 3GB per day assuming a 16 hours a day, capturing 10 frames per second, uploading the cropped images). There are many embedded systems capable of running small neural networks in real-time, however, it is still important to validate that our RLN can run on-device processing 10 frames per second in real-time. In terms of the camera hardware, miniature cameras⁹ that can easily fit inside a typical spectacle frame exist, still capturing images at the same resolution we used (640 × 480). We believe that these cameras as is will work with our system since object recognition in deep learning is basically robust for different cameras. Therefore making a high fidelity prototype with these cameras to will enable realization of a practical, wearable device.

8. Conclusion

In this article, we developed a technique to automatically analyze the corneal surface reflections collected from a single RGB camera. Our technique enable the development of passive lifelogging applications without having to calibrate for individual users. We collected unlabeled and labeled data to develop a two stage synthetic and self-supervised learning based pipeline to automatically detect and crop the reflective area and recognize the focused object. We pretrained the cropping network by developing a synthetic image generation function and later finetuned on a small set of annotated data. To train the object recognition algorithm, we used a contrastive training framework called SimCLR, and finetuned on a smaller set of annotated data. Our results show that corneal surface reflections can be used as a person-independent source of information for passive lifelogging. We provide a discussion on feasibility, challenges and opportunities to develop lifelogging applications based on corneal surface reflections. We believe that our work provides a good starting ground for future research for robust and generalizable lifelogging wearable devices that uses corneal surface reflections from a single wearable camera.

Notes

1. <https://docs.pupil-labs.com/#diy>
2. Access to the dataset is granted upon request to authors via tharindu@ahlab.org
3. https://drive.google.com/file/d/1jTOMjkd8jTEbrxf/_6sVbtA-VG5bsLlhv/view?usp=sharing, <https://drive.google.com/file/d/11ugbXqMZQVC1SwoUHF1pzeNtzY4WM64y/view?usp=sharing>
4. <https://docs.pupil-labs.com/#diy>
5. <https://www.amazon.com/Microsoft-LifeCam-HD-6000-Webcam-Notebooks/dp/B00372567A>
6. <https://github.com/google-research/simclr>
7. <https://www.tobiipro.com/product-listing/tobii-pro-glasses-3/>
8. <https://design.google/library/ux-ai/>
9. <https://www.digikay.com/en/products/detail/omnivision-tech-nologies-inc/OVM7692-RAAA/3190565>

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